MINI PROJECT REPORT

Student Performance Analytics System

Python • NumPy • Pandas

Submitted by	Rohan Mathur
Project type	Python with AI — Mini Project
Dataset size	20 student records

GitHub Repository

https://github.com/rohanmathur-ctrl/student-performance-analytics

1. Project Objective

The objective of this project is to build a simple student performance analysis system using Python. It reads student data from a CSV file, calculates marks and averages, assigns grades, determines pass/fail status, compares subject performance, and identifies the top-performing students.

2. Technologies Used

Python, NumPy, Pandas, and Google Colab.

3. Dataset Description

The CSV dataset contains 20 student records with the following fields: Student_ID, Name, Department, Maths, Python, AI, and Attendance. The three subject marks are used for total, average, grade, and result calculations.

4. Implementation

• Imported NumPy and Pandas.

• Loaded the students.csv file into a Pandas DataFrame.

• Calculated each student's total marks and average across Maths, Python, and AI.

• Used a reusable function to assign grades.

• Used a condition to classify students as Pass or Fail.

• Calculated class statistics and subject-wise averages.

• Identified the best-performing subject and top three students.

• Ran basic assertions to test the calculations and results.

5. Grading and Pass/Fail Rules

Average Marks	Grade
90 and above	A
80 to below 90	B
70 to below 80	C
60 to below 70	D
Below 60	F

Pass rule: A student passes only if they score at least 40 marks in each of the three subjects.

6. Results and Output

Metric	Result
Total students	20
Class average	73.82
Highest average	96.00
Lowest average	37.67
Best performer	Isha (average: 96.00)
Subject averages	Maths: 72.00 | Python: 74.80 | AI: 74.65
Best-performing subject	Python
Students passed	19
Students failed	1
Top 3 students	Isha (96.00), Neha (94.33), Anjali (91.67)

7. Project Files

• students.csv — student dataset

• Student_Performance_Analytics.ipynb — project code and analysis

• README.md — repository overview and run instructions

• Project documentation PDF — this report

8. Challenges and Learning

This project provided practice in loading CSV data, working with DataFrames, performing numerical calculations with NumPy and Pandas, writing reusable functions, using loops and conditions, and validating outputs with tests.

9. Conclusion

The Student Performance Analytics System successfully processes the dataset and produces useful summaries of student marks, grades, pass/fail status, subject averages, and top performers. It demonstrates the core Python

programming and data analysis concepts required for this mini project.

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